Blessed G/Michael Catholic Secondary School

2nd Semester Chemistry test for Grade 10th

Time Allowed: - 1; 20hr

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Name Betanya Wondemagegnehu **NO** 12 **Section** 10th B

I. Choose the best answer from the given alternatives.

D 1. Which of the following is an amphoteric oxide?

A. P_4O_{10} B. N_2O_2 C. CaO D. SnO_2

A 2. According to the Bronsted-Lowery definitions, an acid is a substance that

A. Donates proton to a base

B. Accepts electron pair from a base

C. Donates electron pair to a base

D. Accepts proton from a base

B 3. What is the PH of a solution that has a hydrogen ion concentration 100 times greater than a solution with a P^H of 4?

A.3 B.2 C. 5 D.6

B 4. A certain base that was purchased from a super market ionizes almost completely in water. Then, it is a

A. Dilute base B .Strong base C. Concentrated base D. Weak base

C 5. Which of the following oxides is not salt- forming?

A. Basic oxides B. Amphoteric oxides C. Neutral oxides D. Acidic oxides

C 6. Which of the following pair of oxides is amphoteric?

A. SnO, and H_2O_2 B. PbO, and CaO_2 C. ZnO, and PbO D. Al_2O_3, and Na_2O_2

B 7. Which of the following represents a ternary acid?

A. HF B. $HClO_4$ C. HCl D. HCN

B 8. The strength of an acid can be determined by

A. The number of oxygen atoms

B. Its degree of ionization in water

C. Its molar concentration

D. The number 0f hydrogen atom

C 9. Among the following, identify a substance that is capable of donating a pair of electrons.

A. A Lewis acid. B. An Arrhenius acid. C. A Lewis base. D. An Arrhenius base.

D 10. The base that is used as an electrolyte in some dry cells as well as to make soft soaps is

A. Calcium hydroxide B. Sodium hydroxide C. Ammonia D. Potassium hydroxide

A 11. Which of the following set of compounds are salts?

A. $MgCl_2$, Na_2CO_3.$CaSO_4$

B. NaOH, $Mg(OH)_2$, $Fe(OH)_3$

C. HBr, HNO_3, H_2SO_4

D. MgO, Fe_2O_3, NO

D 12. Binary compounds that contain oxygen and any other element are

A. Acids B. Salts C. Bases D. Oxides

A 13. In a certain experiment KOH solution is added from a burette is added from a burette to dilute hydrochloric acid in a beaker. Which of the following changes will occur in the beaker?

A. Concentration of hydroxide ion decreases

B. Concentration of hydrogen ion increases

C. P^{OH} increases

D. P^H increases

A 14. According to Lewis definition, an acid is

A.an electron pair acceptor

B. A substance that release hydrogen ion or proton, in solution

C.an electron pair donor

D. A substance which dissociates in aqueous solution to release hydroxide ions

A 15. Which of the following substances is a Lewis acid?

A. BF_3 B.H_2O C.NH_3 D.PH_3

A 16. Which of the following reactions does not give a base?

A. $SO_2 + H2O \rightarrow H_2SO_3$ (Acid)

B. $Na_2O + H_2O \rightarrow NaOH$

C.$K_2SO_4 + Ba(OH)_2 \rightarrow$

D.$Ca + H_2O \rightarrow$

A 17. Which of the following salts is used to prevent fungal attack of leaves and vines?

A.$CuSO_4$ B.$FeCl_3$ C. NaCl D.$CaSO_4$ → gypsum

C 18. Which of the following is a strong electrolytic solution? 0.1M solution of

A. Carbonic acid B. Acetic acid C. Potassium hydroxide D. Ammonia

A C 19. Which of the following reaction of chlorine gives an acidic solution?

A. $Cl_2(g) + H_2O(l) \rightarrow Cl^- + H^+$

B. $Cl_2(g) + 2Br\text{-}(aq) \rightarrow$

C. $H_2(g) + Cl_2(g) \rightarrow$

D.$2Fe(s) + 3Cl_2(g) \rightarrow$

C 20. Which of the following salts can be classified as acidic?

A. Zn (OH) Cl → basic B. $MgSO_4$ → N C. $NaHSO_4$ → A D. KCl → N

C 21. The table below shows typical PH values found in four parts of the human digestive system?

Part of human Digestive system	PH
Mouth	6.7
Stomach	2.0
Pancreas	8.0
Small intestine	7.3

High

Basic

neutral

Which part of the digestive system has the least acidic environment?

A. Stomach B. Small intestine C. Mouth D. Pancreas

C 22. Given the following table that contains information about three solutions;

Which list has the solutions placed in order of increasing hydrogen ion concentration?

Solution	PH value
1	10
2	7
3	5

$[H^+] = 10^{-PH} = 10^{-10}$

10^{-7}

10^{-5}

A. 3,1,2 B. 3,2,1 C. 1,2,3 D.2,1,3

D B (23) All of the following can act as Bronsted –Lowery bases except.

A.I^- B.NH_3 C. HCO_3^- D.H_2CO_3

B 24. Which species is amphiprotic?

A H_2S B HS^- C Cl^- D HCl

25. Which species is a Lewis acid in the reaction? $FeCl_3 + Cl^- \rightarrow FeCl_4^-$

A. $FeCl_3$ B. Cl- C. $FeCl_4^-$ D. None

26. Which of the following solution is basic?

A. $[H^+]$ $2*10^{-5}$ M B. $[OH^-] = 3*10^{-9}$ M C. $[OH^-] = 1*10^{-7}$ M D. $[H^+] = 1*10^{-8}$M

27. A solution which has a PH of 5.0

A. Is basic B. Has $[H^+] = 10^{-5}$M C. Has $[H^+] = 10^{-9}$M D. Is neutral

28. What is the hydroxide ion concentration for a solution with a P^H of 10?

A. 10^{-14}M B. 10^{-10}M C. 10^{-7}M D. 10^{-4}M

29. Which of the following is the weakest Bronsted- Lowery acid?

A. H_2SO_4 B. HNO_3 C. HF D. HI

30. A solution with a P^H of 7.5 would be described as;

A. Very basic B. Slightly basic C. Slightly acidic D. Very acidic

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1. D	11. A	21. C
2. A	12. D	22. C
3. B	13. A	23. D
4. B	14. A	24. B
5. C	15. A	25. A
6. C	16. A	26. D
7. B	17. A	27. B
8. B	18. C	28. D
9. C	19. C	29. C
10. D	20. C	30. B

BONUS (Show all necessary steps)

What is the P^{OH} of a 500 ml aqueous solution of 2 gram of NaOH? = 1

(Atomic masses; Na= 23, O=16, H=1)

$NaOH = Na^+ + OH^-$

Mole of NaOH = $\frac{\text{g.mass}}{\text{calculated mass}} = \frac{2g}{40g} = 0.05$ mole

Calculate Molarity: $\frac{\text{mole}}{\text{volume}} = \frac{0.05}{0.5} = 0.1M$

$0.1\,NaOH = Na^+ + \alpha\,OH^-$

$\alpha = 0.1M \times 1 = 0.1M = OH^-$

$POH = -\log[OH^-]$

$-\log(1 \times 10^{-1})$

$-(-1) = POH = 1$

change 500 ml to liters

500 ml $\Rightarrow \alpha$

1000 ml = 1 l

$\frac{1000\,ml \times \alpha}{1000} = \frac{500\,ml}{1000}$

$\alpha = 0.5$

29

30

Good